Firewalls Basics

Network Traffic Control and Perimeter Defense

Firewalls act as the primary security guards of a digital environment, monitoring and controlling incoming and outgoing traffic based on predetermined security rules. This module explores the evolution of firewall technology, from basic **Stateless** and **Stateful** filtering to advanced **Proxy** gateways and **Next-Generation Firewalls (NGFW)** that perform deep packet inspection. We examine the fundamental components of a firewall rule, including **Source/Destination addresses**, **Ports**, and **Protocols**.
Furthermore, the content provides hands-on guidance for managing the **Windows Defender Firewall** through custom rules and network profiles, as well as utilizing Linux utilities like **ufw** (Uncomplicated Firewall) to secure terminal-based systems. Mastering these tools is essential for reducing the attack surface and preventing unauthorized network intrusions.

This **documentation** and **learning** approach is strongly inspired by the structured, hands-on content provided by **TryHackMe**, which significantly contributed to practical understanding of "Firewall basics" concepts.

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1. Purpose and Function of a Firewall

A firewall is a security solution designed to inspect a network's or digital device's traffic. It acts as a barrier between a trusted internal network and untrusted external networks (like the Internet).

- **Rule-Based Filtering:** The firewall compares every piece of data against a list of rules to decide whether to **Allow** or **Deny** access.
- **Perimeter Defense:** It ensures that only authorized visitors and data packets can enter or leave the system.

2. Types of Firewalls

Firewalls operate at different layers of the **OSI Model**, providing varying levels of security and performance.

Firewall Type	OSI Layer	Key Characteristics
Stateless	3 & 4	Fast; filters packets individually; does not remember previous connections.
Stateful	3 & 4	Tracks the "state" of connections in a **State Table**; more secure than stateless.
Proxy	7	Intermediary gateway; inspects application-level data and provides anonymity.
NGFW	3 - 7	Most advanced; includes **IPS**, heuristic analysis, and **SSL/TLS Decryption**.

3. Firewall Rules and Components

Rules are the instructions that tell the firewall how to handle traffic. A standard rule consists of specific criteria and a resulting action.

Core Rule Components

- **Source Address:** The IP originating the traffic.
- **Destination Address:** The IP receiving the data.
- **Port:** The communication "doorway" (e.g., 80 for HTTP).
- **Protocol:** The transport method (TCP or UDP).
- **Direction:** Whether the rule applies to **Inbound** (incoming) or **Outbound** (outgoing) traffic.
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Standard Actions

1. **Allow:** Permits the traffic to pass through.
2. **Deny:** Blocks the traffic and prevents access.
3. **Forward:** Redirects the traffic to a different internal network segment.

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4. Windows Defender Firewall

Windows includes a built-in firewall that utilizes **Network Profiles** to apply settings based on the user's location.

- **Private Networks:** Applied when connected to trusted home or office networks.
- **Public Networks:** Applied in untrusted areas (e.g., coffee shops) to block all non-essential incoming connections.

Creating Custom Rules

Through **Advanced Settings**, users can create specific rules to block applications or ports.

- **Example:** To block web browsing, a user would create an **Outbound Rule** for **TCP Ports 80 and 443**.

5. Linux Firewalls (Netfilter & UFW)

Linux security relies on the **Netfilter** framework. While **iptables** is the traditional tool, **ufw** (Uncomplicated Firewall) provides a simplified interface for beginners.

Command Section: UFW Basics

Task	Command	Description
Check Status	sudo ufw status	Shows if the firewall is active.
Enable	sudo ufw enable	Starts the firewall.
Default Deny	ufw default deny outgoing	Blocks all traffic leaving the machine by default.
Block Port	sudo ufw deny 22/tcp	Blocks all incoming **SSH** traffic.
Delete Rule	sudo ufw delete [number]	Removes a rule based on its ID.

6. Practical Tasks: Windows Rules Analysis

I researched the Windows Defender configuration on a critical system and **found** several custom rules implemented by the security team. Below is how **I solved** the analysis:

- **Work Done: I practiced** navigating the **Advanced Security** console to identify active Inbound and Outbound restrictions.
- **Target Rule 1: I found** a rule named "Block SSH" which was explicitly created to deny all traffic on **TCP Port 22**.
- **Target Rule 2: I researched** the exceptions and **found** a rule named "Allow SSH from Admin" that permitted traffic from a specific IP.
- **Result: I founded** that the allowed IP address under this specific rule was **10.10.10.10**.

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Conclusion and Cheatsheet

Conclusion

Firewalls are a fundamental layer of defense in any security architecture. Whether using the graphical interface of **Windows Defender** or the command-line simplicity of **ufw**, the goal remains the same: restricting the environment to "least privilege" access. By understanding the difference between **Stateful** inspection and **Next-Gen** heuristic analysis, security professionals can deploy the right level of protection to match the organization's threat profile.

Fast Revision Cheatsheet

- **Inbound:** Traffic coming **into** your network.
- **Outbound:** Traffic **leaving** your network.
- **Port 80/443:** Standard ports for Web (HTTP/HTTPS).
- **Port 22:** Standard port for Remote Access (SSH).
- **Stateful:** Remember connections; Stateless: Forget connections.
- **NGFW:** Includes **Deep Packet Inspection** and **IPS**.
- **UFW:** The "Uncomplicated" way to manage Linux firewall rules.
- **Action Deny:** Stops traffic immediately.

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